MA'ADEN معادن

Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project **Mansourah Massarah Gold Project**

Customer: **MA'ADEN GOLD AND BASE METALS (MA'ADEN)**

Title: **DESIGN CALCULATION**
3370 FLOTATION PROCESS WATER STORAGE AND DISTRIBUTION
DESIGN OF TANK FOUNDATION

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DEFINITIONS AND ABBREVIATIONS 3

1. INTENT 4

2. SCOPE 4

3. DESIGN REQUIREMENTS

3.1 Design Loads 4

4. CONCRETE FOUNDATIONS

4.1 Design 4

4.2 Materials 4

4.3 Foundations 5

4.4 Minimum Concrete Cover 5

4.5 Allowable Soil Bearing Capacity 5

5. DESIGN CALCULATION FOR FOUNDATION 6

References 11

ATTACHMENT 1 LOAD DATA DRAWINGS 12

ATTACHMENT 2 COMMENT RESOLUTION SHEET 13

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DEFINITIONS AND ABBREVIATIONS

Employer	MA'ADEN GOLD AND BASE METALS or their representatives
EPC Contractor	Consortium of Outotec (OT) and Larsen & Toubro (L&T)
Project management Company (PMC)	BECHTEL appointed by EMPLOYER
Supplier	An entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Sub- supplier	A sub entity (supplier, consultant, contractor) under the direct project management of EPC Contractor
Shall	Is to be understood as mandatory in relation to the requirements of this Document
Should	Is to be understood as a strong recommendation to comply with the requirements of this Document

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1. INTENT

This document is to prepare design of **Flotation Process Water (3370)** Tank. The analysis & design is carried out in accordance with project specifications and codes & standards. The design calculation ensures the safety and serviceability of the foundation system.

2. SCOPE

The analysis has been carried out in Excel. For member strength calculation force values are directly taken from Structural Calculation & designed in excel sheet.

3. DESIGN REQUIREMENTS

3.1 Design Load

All Design Loads are indicated in Load Data (Drawing No. MD-216-3370-EG-ST-GAD-1001) attached in Appendix A. Also referred Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.2.

4. CONCRETE FOUNDATIONS

4.1 Design

Design of concrete structures and foundations shall be in accordance with SBC-304, ACI 318M and API 650.

4.2 Materials

Concrete Strength of 28 days

- Foundation and Structural Members: 35 MPa
- Grade slab and pavement: 30 MPa
- Blinding Concrete below foundation bases: 15 MPa

Reinforcement Grade

- Deformed Bar : ASTM A 615 / A 615M Grade 60 (fy = 420MPa)
- Welded Wire Fabric: ASTM A 185(Plain) / A 497(Deformed)

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4.3 Foundations

Foundation shall be Earth Foundation with a Concrete Ring wall for Tanks. The design calculations are prepared as per recommendation of standard API 650 - Annex B.

4.4 Minimum Concrete Cover

Concrete Exposure	Member	Minimum Cover(mm)
Exposed to weather or contact with ground	Wall	50

4.5 Allowable Soil Bearing Capacity

The net safe bearing capacity of soil for foundations design has been considered 37.5 kN/ m^2 at 0.6 m below EGL (ie. RL 911.40 m) refer Table 5 - BH-26 of Geotechnical Report (MD-216-3000-EG-CV-RPT-1008).

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5. DESIGN CALCULATION FOR FOUNDATION

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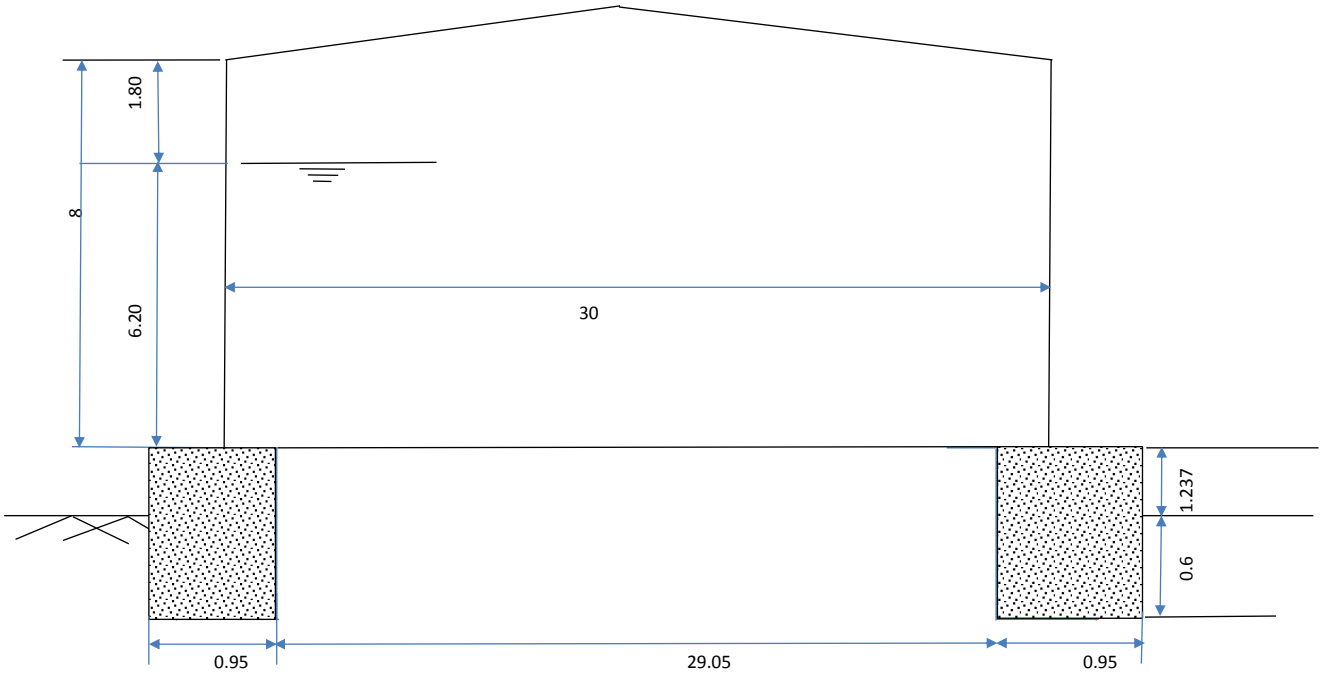
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Tank Loading Data

	Loading Type		
Weight, V	Empty (W)	1795	kN
	Operating (Wo)	45516	kN
	Test (Wt)	57269	kN

Summary of Tank Specification



Diameter of Tank, D	=	30 m	
Height of Tank, H	=	8 m	
Height of Tank Fluid, H_L	=	6.2 m	
Inner Diameter of Concrete Ring Wall, ID	=	29.05 m	
Outer Diameter of Concrete Ring Wall, OD	=	30.95 m	
Height of Concrete Ring Wall, h	=	1.837 m	
Thickness of Concrete Ring Wall, t	=	0.95 m	
B.C.D	=	27.752 m	
Top of Asphalt Board	=	1.237 m	(Above Finished Level)
Unit Weight of Liquid, γ_L	=	9.95 kN/m^3	
Unit Weight of Soil, γ_S	=	18 kN/m^3	
Unit Weight of Concrete, γ_C	=	24 kN/m^3	

Wind Load & Seismic Load

Load	Load Condition	Base Shear (F,kN)	Moment (M,kN-m)
Wind Loads	Empty	129.00	531.00
	Operating	129.00	531.00
Seismic Loads	Operating	217.00	582.00

Foundation Weight

a) Ring Wall Weight = 3947.44 kN

a) Inside Soil Weight = 21916.13 kN

Sectional Area

Net cross-sectional area for empty condition, Ae = 89.54 m^2

Full cross-sectional area for operating condition, Ao= 752.33 m^2

Sectional Modulus

Sectional modulus for empty condition, Ze = 651.56 m^3

Sectional modulus for operating condition,Zo = 2910.60 m^3

Check of Minimum Wall Thickness (As per API 650 ANNEX-B)

As the differential settlement is disadvantageous to tank,minimum thickness of concrete wall required to equalize soil pressures between under tank and ringwall will be checked.

Soil pressure under the tank, $P = \gamma_L * H_L + \gamma_s * h$

Soil pressure under the ringwall, $P = w / t + \gamma_L H_L / 2 + \gamma c\ h$

Solving the above 2 equations, $\gamma_L * H_L + \gamma_s * h = w / t + \gamma_L H_L / 2 + \gamma c\ h$

or, $t = 2w / (\gamma_L * H_L - 2h(\gamma_c - \gamma s))$

w = Weight of ring tank shell plus tributary weight of roof per meter circumference, $(W/\pi D)$ = 19.05 kN/m

So, Required Min Wall THK.= 0.9608 m = 960.8 mm > 950 mm NOT OK

Basic Load Cases

1. Tank Dead Load (D)
2. Operational Fluid Load (F1)
3. Test/Max Fluid Load (F2)
4. Wind Load (W)
5. Seismic Load (E)

Allowable Stress Design

Load Combination (As per SBC-301 ,2007, Cl No. 2.4.1)

Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	H (kN)	M (kNm)
Long term	Normal Operating	LC101	D + F1	45516.00	0.00	0.00
Short term	Empty	LC102	0.6 D + W	1077.00	129.00	531.00
	Operating	LC103	D + F1 + W	45516.00	129.00	531.00
		LC104	D + F1+ 0.7 E	45516.00	151.90	407.40
		LC105	0.6 (D + F1) + 0.7 E	27309.60	151.90	407.40
	Test	LC106	D + F2	57269.00	0.00	0.00

Stability Check

Allowable Bearing Capacity of Soil, Q_{all} = 375 kN/m^2

An increase in allowable bearing capacities by about 33 percent may be considered under short term loading conditions like wind , earthquake.

Q_{all} = 498.75 KN/m^2

Load Case	Loads at bottom of foundation level		Qmax= V/A + M/Z kN/m^2	Qmax= V/A - M/Z kN/m^2	Qall/ Qmax	Soil Bearing Check
	V (kN)	M(kNm)				
LC101	71379.56	0.00	94.88	94.88	3.95	OK
LC102	5024.44	767.97	57.30	54.94	8.70	OK
LC103	71379.56	767.97	95.14	94.61	5.24	OK
LC104	71379.56	686.44	95.11	94.64	5.24	OK
LC105	53173.16	686.44	70.91	70.44	7.03	OK
LC106	83132.56	0.00	110.50	110.50	3.39	OK

Sliding Check

Calculate safety factor against sliding by dividing resisting shear by actual shear

Fs =H resistance force / H acting force > 1.5

Load Case	Loads at bottom of foundation level	Coef. of friction	H_{resist} (kN)	H_{acting} (kN)	Fs	Sliding Check
	V (kN)					
LC101	71379.56	0.4	28551.83	0.00	-	-
LC102	5024.44	0.4	2009.77	129.00	15.6	OK
LC103	71379.56	0.4	28551.83	129.00	221.3	OK
LC104	71379.56	0.4	28551.83	151.90	188.0	OK
LC105	53173.16	0.4	21269.27	151.90	140.0	OK
LC106	83132.56	0.4	33253.03	0.00	-	-

Overturning Check

Calculate safety factor against overturning by dividing resisting moment by overturning moment

The resultant Safety Factor should be 1.5 minimum

Fs =M resistance moment / M overturning moment > 1.5

Load Case	Loads at bottom of foundation level	Radius of fdn.	M_{resist} (kNm)	M_{acting} (kNm)	Fs	Overturning Check
	V (kN)					
LC101	71379.56	15.475	1104598.75	0.00	-	-
LC102	5024.44	15.475	77753.15	767.97	101.2	OK
LC103	71379.56	15.475	1104598.75	767.97	1438.3	OK
LC104	71379.56	15.475	1104598.75	686.44	1609.2	OK
LC105	53173.16	15.475	822854.71	686.44	1198.7	OK
LC106	83132.56	15.475	1286476.42	0.00	-	-

Ultimate Strength Design

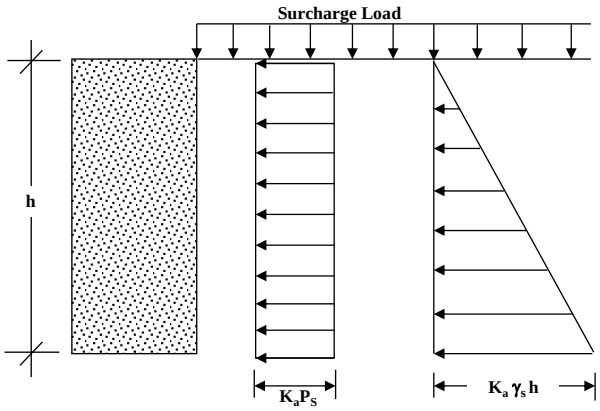
Load Combination (As per SBC-304 , 2018,Table 5.3.1)

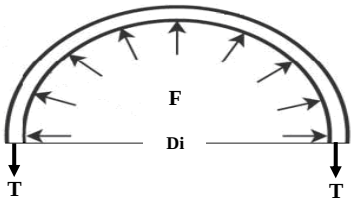
Term of Load	Load Conditions	Load Case	Load Combination	V (kN)	M (kNm)	A (m^2)	Z (m^3)
Long term	Normal Operating	LC101	1.4 (D + F1)	63722.40	0.00	706.86	2650.72
Short term	Empty	LC102	0.9 D + 1.6W	1615.50	849.60	706.86	2650.72
	Operating	LC103	1.2 (D + F1) +1.6 W	54619.20	849.60	706.86	2650.72
		LC104	1.2 (D + F1) + E	54619.20	582.00	706.86	2650.72
		LC105	0.9 (D + F1) + E	40964.40	582.00	706.86	2650.72
	Test	LC106	1.4 (D + F2)	80176.60	0.00	706.86	2650.72

Surcharge Load

Load Case	Loads at bottom of Tank level		Surcharge= V/A + M/Z
	V (kN)	M(kNm)	kN/m^2
LC101	63722.40	0.00	90.15
LC102	1615.50	849.60	2.61
LC103	54619.20	849.60	77.59
LC104	54619.20	582.00	77.49
LC105	40964.40	582.00	58.17
LC106	80176.60	0.00	113.43

Maximum Surcharge Load, Ps = 113.43 kN/m^2





Where, K_a = coefficient of earth pressure at rest
= 0.5

P1 = $K_a * P_s$ = 56.71 kN/m^2

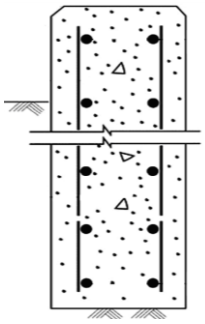
P2 = $1.4 * \gamma_S * h * K_a$ = 23.15 kN/m^2 for LC106 1.4 (D + F2)

Maximum Lateral Load

F = P1 * h + 1/2 * P2 * h = 125.442 kN/m

Hoop Tension Load

T= 1/2 * F * ID = 1822.05 kN



Hoop Tension Bar

1. Horizontal Reinforcement

ϕ = 0.9 (As per SBC 304 , Table No. 21.2.2)

Fy = 420 MPa

Required As = T/(φ x Fy) = 4820.23 mm^2

Minimum As = 0.0025 x b x D = 4362.88 mm^2 (As per API 650 , Cl. No. B.4.2.3)

Hence, Provide 28 Nos. D16= 5080.835 mm^2 OK

2. Vertical Reinforcement

Minimum As = 0.0015 * b * D = 1425 mm^2 (As per API 650 , Cl. No. B.4.2.3)

So, Used As = 2 row * 8 ea - D12 (@ 125mm)

=1810mm2 > Required As OK

REFERENCE

Reference is made to codes, standards and specifications as indicated in Civil, Structural and Architectural Design Criteria (Doc. No. MD-216-1000-EG-ST-CRT-1001) Clause 4.1.

ATTACHMENT 1: LOAD DATA DRAWINGS

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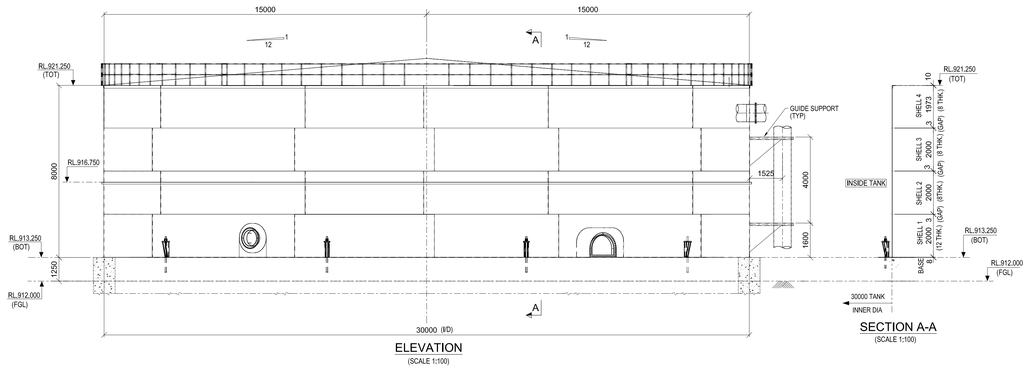
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DRAWING NUMBER: MD - 216 - 3370 - EG - ST - GAD - 1001 | REV: D01

MATERIAL SPECIFICATION	
SHELL, BOTTOM, ANNULAR, ROOF	EN 10025 GR.S275 J0
COD	EN 10025 GR.S275 J0
MANWAY NECKS/ MANWAY FLANGES	EN 10025 GR.S275 J0
NOZZLE NECKS (PIPES) (600 NB & BELOW)	ASTM A53 Gr.B/ ASTM A106 Gr.B
NOZZLE NECKS (PIPES) (> 600 NB)	ASTM A36 GR.36 OR EN 10025 GR.S275 J0/S355 J0
NOZZLE FLANGES/ BLIND FLANGE	SA 105
GASKETS FOR NOZZLES	RING TYPE (FLAT RING) 1.5mm THK, COMPRESSED ASBESTOS FIBRE.
GASKETS FOR MANWAY/ COD	3mm THK, COMPRESSED NON-ASBESTOS SHEETS.
BOLTS/ NUTS FOR NOZZLE	SA 193 Gr. B7 / 194 Gr. 2H
BOLTS/ NUTS FOR MANWAY	SA 193 Gr. B7 / 194 Gr. 2H
STRUCTURAL BOLTS	ASTM A325M
ANCHOR CHAIR	EN 10025 GR.S275 J0
STRUCTURAL(ROLLED SECTION)-STAIRWAY / PLATFORM / ROOF SUPPORTING STRUCTURE	EN 10025 GR.S275 JR
HANDRAIL PIPE / HANDRAIL POST	ASTM A53 TYPE E
PIPE FITTINGS	SA 234 Gr. WPB
REINFORCEMENT PAD/GUSSET/ EXTERNAL SUPPORT ETC.	EN 10025 GR.S275 J0
EARTHING LUGS	EN 10025 GR.S275 J0
ANCHOR BOLTS	ASTM F1554 GR.36
CURB ANGLE	EN 10025 GR.S275 JR
NAME PLATE	STAINLESS STEEL(SS 316)

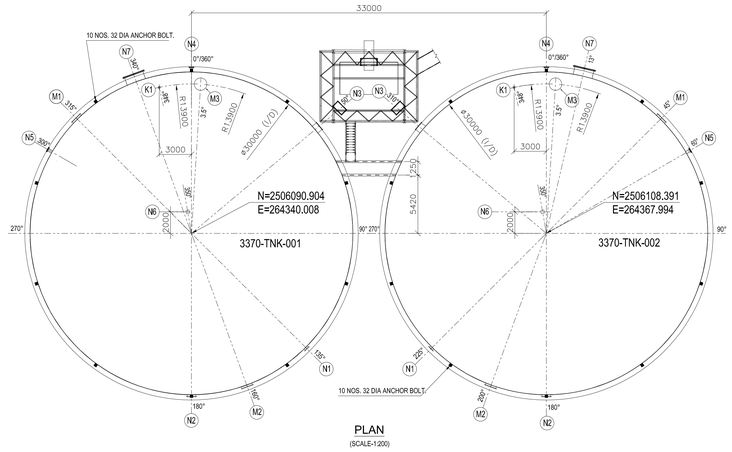
FOUNDATION LOAD DATA:

BASIC LOAD CASE	VERTICAL COMPRESSION (KN)	VERTICAL UPLIFT (KN)	HORIZONTAL SHEAR (KN)	OVERTURNING MOMENT (ABOUT SHELL BOTTOM JOINT) (KN-M)	SLAB MOMENT (KN-M)
DEAD LOAD (NOMINAL THICKNESS)	1795	–	–	–	–
DEAD LOAD (CORRODED THICKNESS)	1598	–	–	–	–
ROOF PIPE LIQUID LOAD	50	–	–	–	–
SHELL PIPE LIQUID LOAD	50	–	–	–	–
LIVE LOAD	924	–	–	–	–
TANK LIQUID LOAD (OPERATING)	43721	–	–	–	–
TANK LIQUID LOAD (TEST)	55474	–	–	–	–
WIND LOAD ON SHELL	–	–	129	531	–
WIND LOAD ON ROOF	–	673	–	10098	–
SEISMIC LOAD	1423	1423	217	582	2423



ELEVATION
(SCALE 1:100)

SECTION A-A
(SCALE 1:100)



PLAN
(SCALE-1:200)

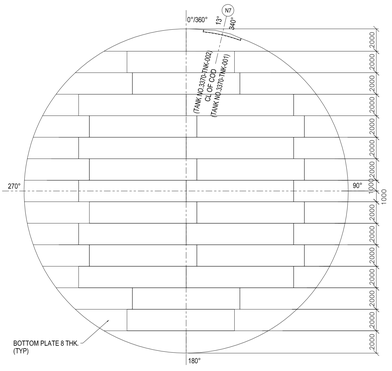
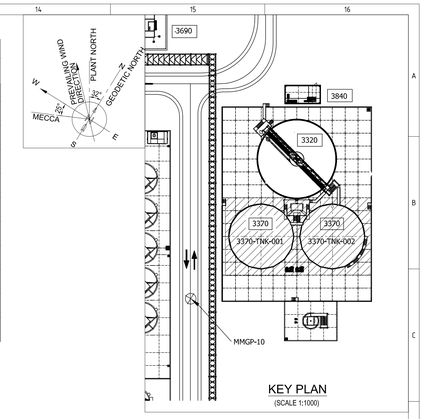


PLATE ARRANGEMENT OF BOTTOM PLATE
(SCALE 1:200)



KEY PLAN
(SCALE 1:1000)

NOZZLE SCHEDULE (FOR TANK NO 3370-TNK-001)

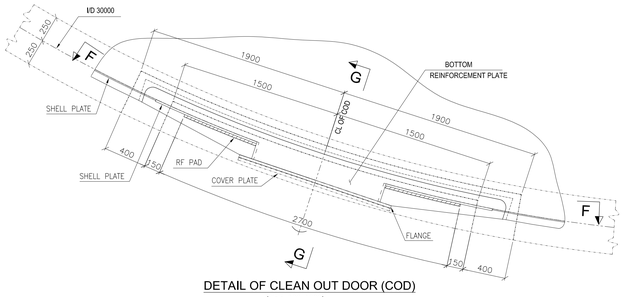
NOZ.	SERVICE	SIZE (DN)	SCH	RATING #	TYPE	ELEV."H"	PROJ."P"	RADIUS "R"	ORIENTATION	LOCATION	RF PAD DIA ★	RF PAD THK. ★	REMARK
N1	TANK OUTLET	500	Sch XS	CL 150	RFSO	750	250	-	135°	SHELL	1055	20	CIRCULAR
N2	OVERFLOW	800	Sch XS	CL 150	RFSO	6712	300	-	180°	SHELL	1645	10	
N3	TANK INLET	800	Sch XS	CL 150	RFSO	7000(TBC)	300	-	50°	SHELL	1645	8	CIRCULAR
N4	DRAIN	100	Sch XS	CL 150	RFSO	260	200	-	0°	SHELL	305	12	LOW TYPE
N5	DRAIN	100	Sch XS	CL 150	RFSO	260	200	-	300°	SHELL	305	12	LOW TYPE
N6	TANK VENT	400x400(TBC)	-	RECTANGULAR WITH COVER ARRANGEMENT		-	300	2000	350°	ROOF			
N7	COD-hxb	900x1200	-	API 650		-	-	-	340°	SHELL	2700x1372	12	
K1	LEVEL TRANSMITTER	100	Sch XS	CL 150	RFSO	-	400	13900	348°	ROOF	305	6	CIRCULAR
M1	MANHOLE C/W DAVIT	750	-	API 650		925	175	-	315°	SHELL	1800	16	LOW TYPE
M2	MANHOLE C/W DAVIT	750	-			925	175	-	160°	SHELL	1800	16	LOW TYPE
M3	MANHOLE C/W HINGED BLIND	600	-			-	500	13900	3.5°	ROOF	1255	6	CIRCULAR

★ REFER NOTES NO. 12

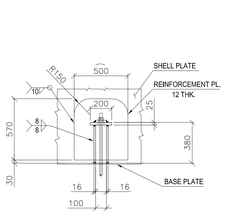
NOZZLE SCHEDULE (FOR TANK NO.3370-TNK-002)

NOZ.	SERVICE	SIZE (DN)	SCH	RATING #	TYPE	ELEV."H"	PROJ."P"	RADIUS "R"	ORIENTATION	LOCATION	RF PAD DIA ★	RF PAD THK. ★	REMARK
N1	TANK OUTLET	500	Sch XS	CL 150	RFSO	750	250	-	225°	SHELL	1055	20	CIRCULAR
N2	OVERFLOW	800	Sch XS	CL 150	RFSO	6712	300	-	180°	SHELL	1645	10	
N3	TANK INLET	800	Sch XS	CL 150	RFSO	7000(TBC)	300	-	310°	SHELL	1645	8	CIRCULAR
N4	DRAIN	100	Sch XS	CL 150	RFSO	260	200	-	0°	SHELL	305	12	LOW TYPE
N5	DRAIN	100	Sch XS	CL 150	RFSO	260	200	-	60°	SHELL	305	12	LOW TYPE
N6	TANK VENT	400x400(TBC)	-	RECTANGULAR WITH COVER ARRANGEMENT		-	300	2000	350°	ROOF			
N7	COD-hxb	900x1200	-	API 650		-	-	-	13°	SHELL	2700x1372	12	
K1	LEVEL TRANSMITTER	100	Sch XS	CL 150	RFSO	-	400	13900	348°	ROOF	305	6	CIRCULAR
M1	MANHOLE C/W DAVIT	750	-	API 650		925	175	-	45°	SHELL	1800	16	LOW TYPE
M2	MANHOLE C/W DAVIT	750	-			925	175	-	200°	SHELL	1800	16	LOW TYPE
M3	MANHOLE C/W HINGED COVER	600	-			-	500	13900	3.5°	ROOF	1255	6	CIRCULAR

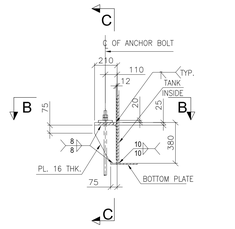
★ REFER NOTES NO. 12



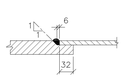
DETAIL OF CLEAN OUT DOOR (COD)
(SIZE 900x1200)
SCALE- 1:20



SECTION C-C
SCALE- 1:20



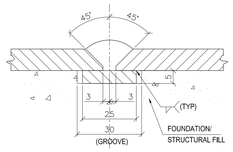
TYP. DETAIL OF ANCHOR CHAIR
SCALE- 1:20



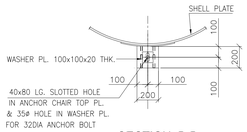
DETAIL-1



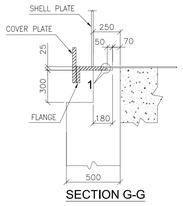
SECTION F-F



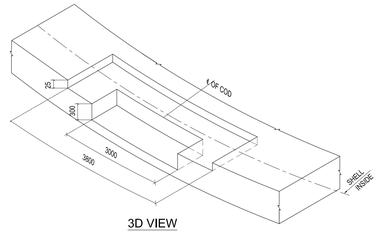
TYP.BOTTOM PLATE JOINT DETAIL



SECTION B-B
SCALE- 1:20



SECTION G-G



3D VIEW

LEGEND:

ID - INNER DIAMETER
OD - OUTER DIAMETER
COD - CLEAN OUT DOOR
LG - LONG
FGL - FINISHED GROUND LEVEL.
BOT - BOTTOM OF TANK
TOT - TOP OF TANK
RL - REDUCED LEVEL
RF - REINFORCEMENT
TOC - TOP OF CONCRETE
GST - GUSSET
PL - PLATE
STIFF - STIFENER
THK - THICKNESS
CLT - CLEAT
UNO - UNLESS NOTED OTHERWISE
BOT - BOTTOM OF TANK
TOT - TOP OF TANK
STORG - STORAGE
DISTBN. - DISTRIBUTION
NOZ. - NOZZLE
DN - DIAMETER NOMINAL
SCH - SCHEDULE
ELEV - ELEVATION
PROJ - PROJECTION
MM - MILLIMETER
TYP. - TYPICAL

DESIGN DATA

TANK DETAILS	TANK NAME		FLOTATION PROCESS WATER TANK 1 & 2
	EQUIPMENT TAG No.		3370-TNK-001 & 002
	TANK DIMENSIONS		DIAMETER : 30000mm, HEIGHT: 8000mm
	QUANTITY		2
	ROOF TYPE		SLOPED CONE ROOF (SLOPE 1 IN 12)
	BOTTOM SLOPE & TYPE		SLOPED BOTTOM (SLOPE 1 IN 120)
	FIELD ERECTED/SHOP FABRICATED		FIELD ERECTED
DESIGN DATA	FLUID		WATER
	DESIGN CODE		API 650-TWELFTH EDITION, MARCH 2013, ADDENDUM 3, AUGUST 2018
OPERATING DATA	OPERATING TEMPERATURE		40° C
	NET WORKING CAPACITY		2806 m^3
	OPERATING LEVEL	MINIMUM	2170 mm
		MAXIMUM	6295 mm
	OPERATING DENSITY OF CONTENT		1.0 t/m^3
DESIGN DATA	DESIGN TEMPERATURE		50° C
	DESIGN PRESSURE (INTERNAL/EXTERNAL)		ATM/ATM
	TEST PRESSURE		ATM
	HYDRO STATIC TEST PRESSURE		FULL OF WATER
	SPECIFIC GRAVITY		1.0
	TESTING		AS PER API 650
	INSPECTION		AS PER API 650
	DENSITY OF SCALE		N/A
	DESIGN SCALE THICKNESS	TANK SHELL/ROOF	N/A
	DESIGN FILL RATE		1144 m^3/hr (TBC)
	DESIGN EMPTY RATE		1144 m^3/hr (TBC)
	CORROSION ALLOWANCE	SHELL	1.6 mm
		ROOF PLATE	0 mm
		BOTTOM PLATE / ANNULAR PLATE	1.6 mm
		ANCHOR BOLT	0 mm
		STRUCTURAL	1.6 mm
WIND LOAD DATA	BASIC WIND SPEED		43 m/sec. (155 Km/hr.)
	SITE EXPOSURE		C
	VELOCITY PRESSURE EXPOSURE COEFFICIENT		K_z = AS PER ELEVATION
	TOPOGRAPHIC FACTOR		K_{zt}=1.0
	WIND DIRECTIONALITY FACTOR		K_d =0.95
SEISMIC LOAD DATA	SITE CLASS		B
	SEISMIC USER GROUP		SUG - I
	IMPORTANCE FACTOR	(I)	I
	SCALING FACTOR	(Q)	2/3
	MAXIMUM DESIGN LIQUID LEVEL	(H)	6.305 m
	CO-EFFICIENT TO ADJUST THE SPECTRAL ACCELERATION FROM 5% TO 0.5% DAMPING	(K)	1.50
	SEISMIC CO-EFFICIENTS		S_s= 0.1g, S_1= 0.02g, F_a = 1.0, F_v= 1.0 S_{ds} = 0.067g, S_{d1} = 0.013g, T = 4 sec.
OTHER DATA	LOCATION		GOLD PROCESSING PLANT
	PWHT REQUIRED(YES/NO)- IF YES,SPECIFY REASON FOR PWHT		YES (NOT ENTIRE TANK)
	REASON FOR PWHT(PROCESS REASON/PER API CODE)		AS PER API 650
	INSULATION	SHELL	N/A
		ROOF	N/A
	RADIOGRAPHY		AS PER CL 8.1 - API 650
	AGITATOR/RAKE REQUIRED (YES/NO)		NO
	EQUIPMENT TYPE (AGITATOR/RAKE)		NA
	AGITATOR/RAKE SUPPORTED BY TANK (YES/NO)		NA
	FOUNDATION TYPE/RING WALL/SLAB)		RING WALL
	LINER REQUIRED (YES/NO)		NO
	LINER MATERIAL		NA
	LINER THICKNESS (ROOF/SHELL/BOTTOM)		NIL
	PAINTING	INTERNAL	YES
		EXTERNAL	YES

NOTES:

1. ALL DIMENSIONS ARE IN MILLIMETER AND LEVELS & COORDINATES ARE IN METER UNO.
2. ANCHOR BOLT MATERIAL SHALL BE AS PER ASTM F 1554 Gr.36,
3. SIZE OF WELD SHALL BE THE LESSER THICKNESS OF THE JOINED MEMBER (UNLESS NOTED OTHERWISE).
4. BOLT HOLES IN NOZZLE FLANGES & MANHOLES SHALL STRADDLE VERTICAL CENTER LINES ON SHELL & NORTH SOUTH CENTERLINES ON ROOF.
5. NOZZLE PROJECTION ON SHELL SIDE REFERS FROM EXTERNAL SURFACE OF SHELL PLATE FACE OF FLANGE. NOZZLE PROJECTION ON ROOF SIDE REFERS FROM TOP OF ROOF PLATE TO FACE OF FLANGE.
6. VERTICAL SHELL JOINTS SHOULD PREFERABLY BE OFFSET BY 1000 MM.
7. CURB ANGLE JOINTS SHALL STAGGER ATLEAST 300 MM WITH VERTICAL SHELL PLATE WELD.
8. REINFORCEMENT PADS TO HAVE TELLTALE HOLES OF 6 MM AND SHALL BE TESTED AT 100 kPa PRESSURE WITH SOAP SOLUTION AND THEN FILLED WITH HARD GREASE.
9. FLANGE (COD & MANHOLE) GASKET SEATING SURFACE SHALL BE PROVIDED WITH 0.0032 TO 0.0063 mm (3.2 TO 6.3 µm) ROUGHNESS FINISH
10. EARTHING LUG DETAIL & LOCATION TO BE CONFIRMED IN SEPARATE DRAWING.
11. SUITABLE NOTCHES AS PER API 650 TO BE PROVIDED IN THE FOUNDATION/GROUT FOR THE COD.
12. MINIMUM THICKNESS OF RF PAD, COLLAR, FLANGE AND COVER PLATES FOR NOZZLES ARE PROVIDED BASED ON HYDROSTATIC LOAD AND FINAL SIZE SHALL BE CONFIRMED AFTER NOZZLE DESIGN IN SEPERATE DRAWING.
13. THIS DRAWING IS TO BE READ IN CONJUNCTION WITH DRAWING NO. MD - 216 - 3370 -EG -ST - GAD - 1002 & 1003

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
MD-216-1000-EG-CV-DED-1003	ANCHOR BOLTS STANDARD DETAILS.								
MD-216-1000-EG-ST-CRT-1001	CIVIL, STRUCTURAL & ARCHITECTURAL - DESIGN CRITERIA	D01	18/11/2023	AS BUILT	SJU	PRI	KDU	AG	
MD-216-3370-EG-ME-GAD-1002	G.A. FLOTATION PROCESS WATER STORAGE TNK-002	C01	07/02/2020	ISSUED FOR CONSTRUCTION AND REVISED AS MARKED	SJU	PRI	KDU	AG	
MD-216-3370-EG-ME-GAD-1001	G.A. FLOTATION PROCESS WATER STORAGE TNK-001	A01	25/12/2019	ISSUED FOR REFERENCE	SJH	PRI	KDU	AG	

RESERVED FOR VENDOR LOGO AND DRAWING NUMBER

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LARSEN & TOUBRO DRAWING NUMBER	019045_450_S_G_26202_001
REVISION	2
OUTOTEC DRAWING NUMBER	900707 __ GAU01 __ AG01 __ MTL01 __ 33704
DOCUMENT ID	OU100667954
REVISION	2
DRAWN	SJH 19/12/2019
CHECKED	PRI 21/12/2019
ENGINEER	KDU 20/12/2019
APPROVED	AG 25/12/2019
EMPLOYER	
SCALE	1:200,100,20
PAPER SIZE	A0

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT DRAWING
FLOTATION PROCESS WATER STORG. & DISTBN. PLAN, ELEVATIONS & DETAILS OF TANK

DRAWING NUMBER	REV
MD - 216 - 3370 - EG - ST - GAD - 1001	D01

Mansourah Massarah Gold Project
Comment Resolution Log Sheet / Compliance Report

PROJECT:	**MANSOURAH MASSARAH GOLD PROJECT** **PROJECT NO.: 216**
STAGE:	**4 - PROJECT EXECUTION.** **REVIEW LOG DOC. NO. & REV.:**
ACONEX REF. NO. & DATE:	BECHTEL-WTRAN-001456, 27 February 2020

STATUS CODE: A - APPROVED, C - APPROVED WITH COMMENTS, R - REVISE & RESUBMIT & O - OTHERS (INFORMATION, REFERENCE, REVIEWED)

DOCUMENT NUMBER	Area / Title	Comment No.	PMC/PMT COMMENT DATE: 27 February 2020	Comment by	EPC Response DATE: 11 March 2020	Status	Remarks
			General Comments (cc)				
MD-216-3370-EG-CV-CAL-1234-A01	3370	1	General Comment Calculation are assuming that seismic forces are of the type LRFD. Please confirm, otherwise calculations need to be adjusted. We have asked for this information but have not received this information from EPC	Mr. Doron Golan BECHTEL	All load combinations modified according to discussion held among PMC, OTE & L&T.	Closed	